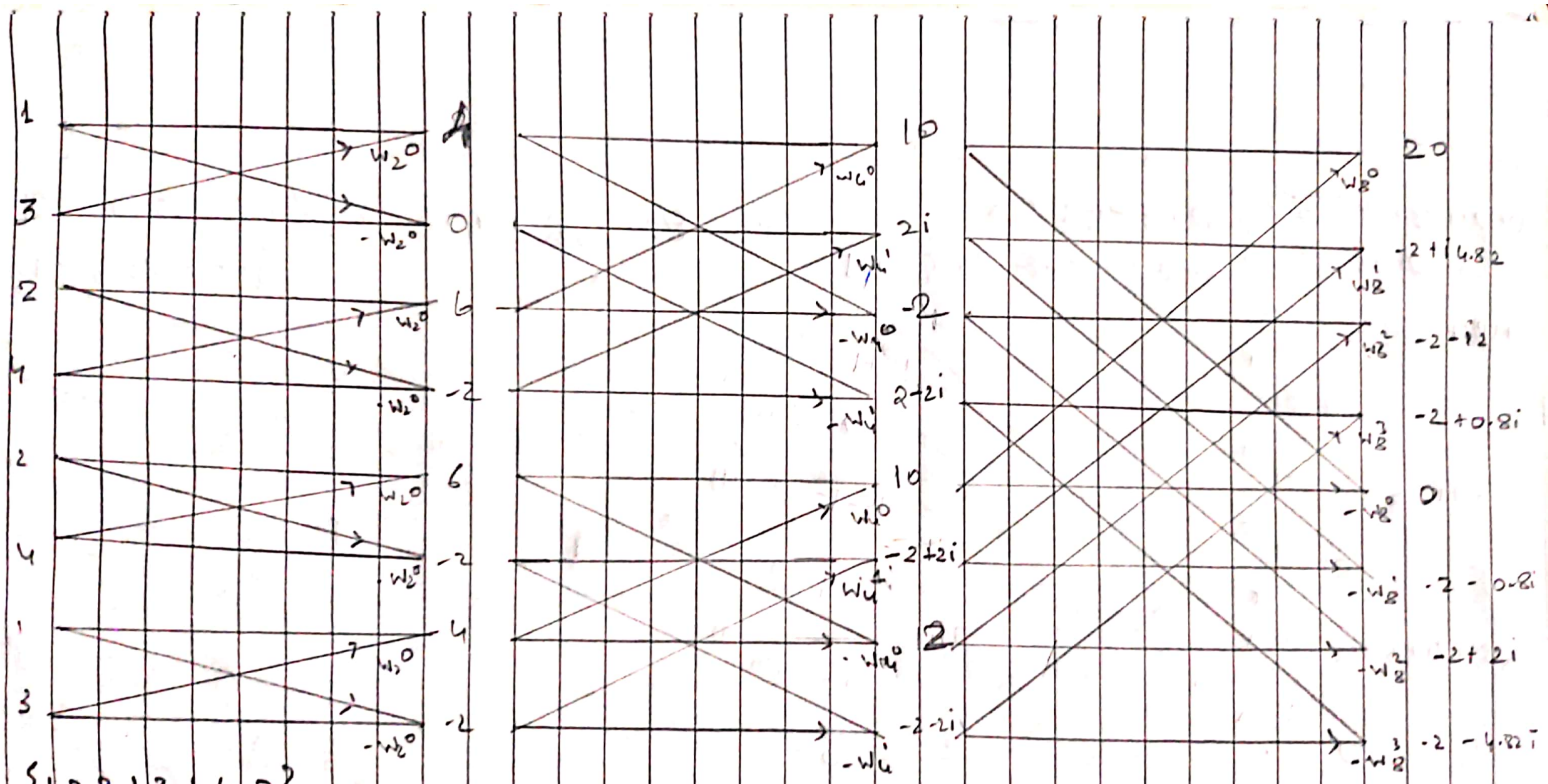

{1 2 2 1 3 4 4 3}

{1 2 3 4}

{2 1 4 3}

{1,3} {2,4} {2,4} {1,3}

FFT = {20, -2 + 4.8i, -2 -2i, -2+0.8i, 0, -2 -0.8i, -2 +2i, -2 -4.82i}

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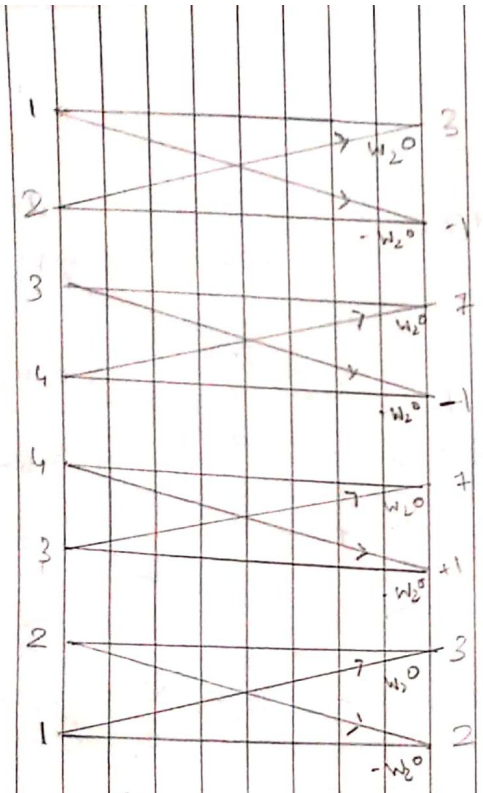
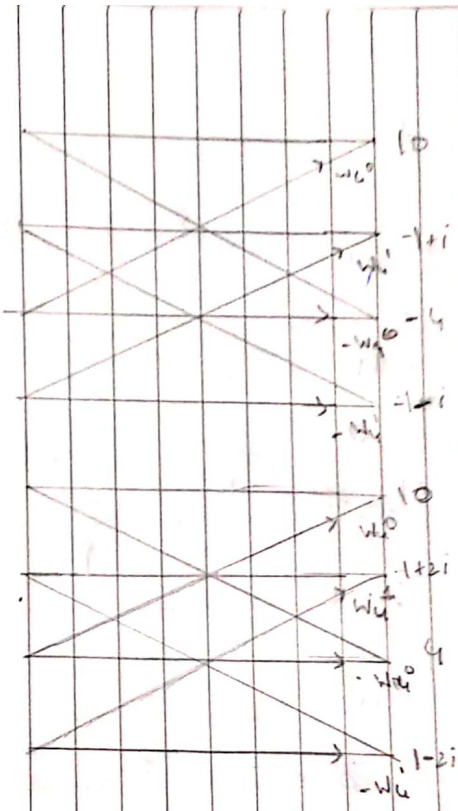
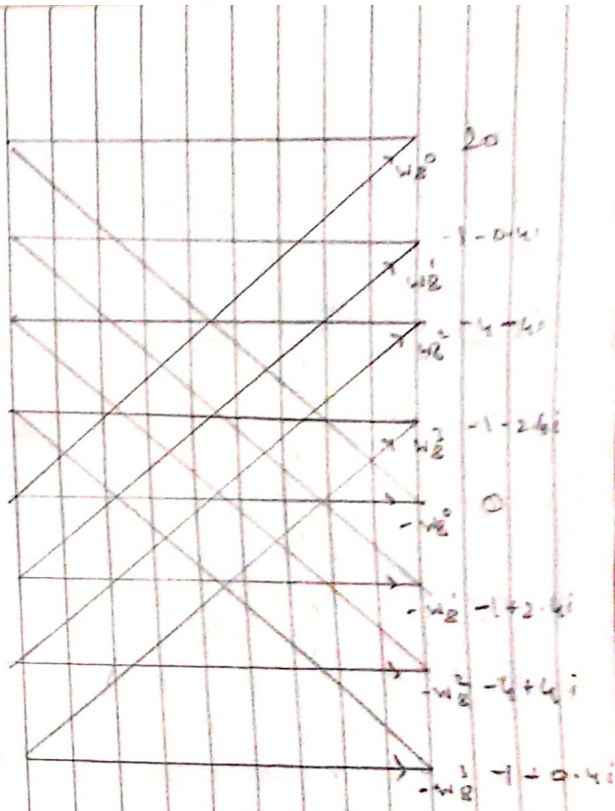






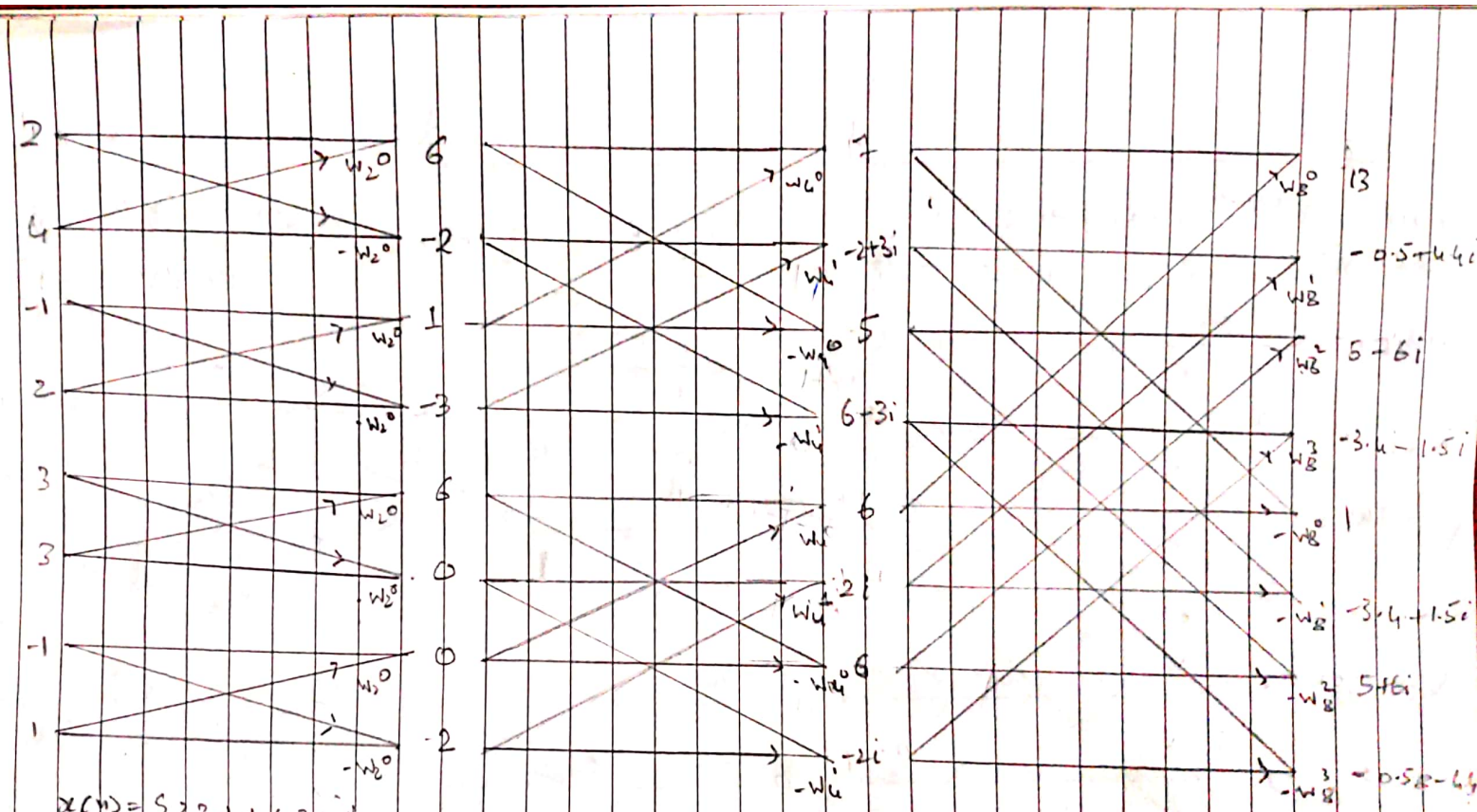
$x(n) = \{1, 4, 3, 2, 2, 3, 4, 1\}$

$\{1, 3, 2, 4\}$ $\{4, 2, 3, 1\}$

$\{1, 2\}$, $\{3, 4\}$ $\{4, 3\}$ $\{2, 1\}$

$FFT = \{20, -1-0.4i, -4-4i, -1-2.4i, 0, -1+2.4i, -4+4i, -1+0.4i\}$

Scanned with CamScanner



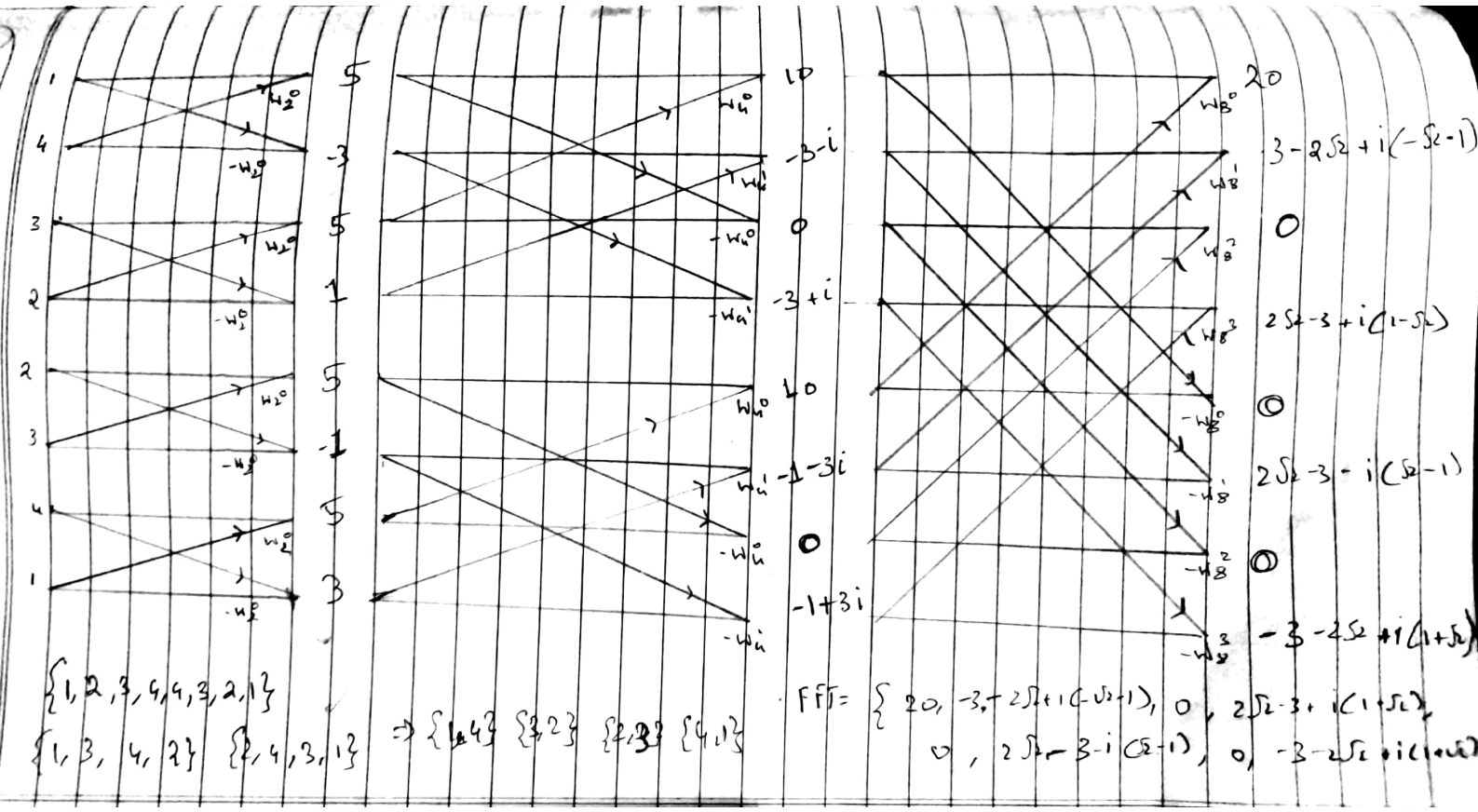
$x(n) = \{2, 3, -1, -1, 4, 3, 2, 1\}$

$\{2, -1, 4, 2\}$ $\{2, 4\}$ $\{-1, 2\}$ $\{3, 3\}$ $\{-1, 1\}$

$\{3, -1, 3, 1\}$

$FFT = \{13, -0.5+4.4i, 5-6i, -3.4-1.5i, 1, -3.4+1.5i, 5+6i, -0.5-4.4i\}$

Scanned with CamScanner

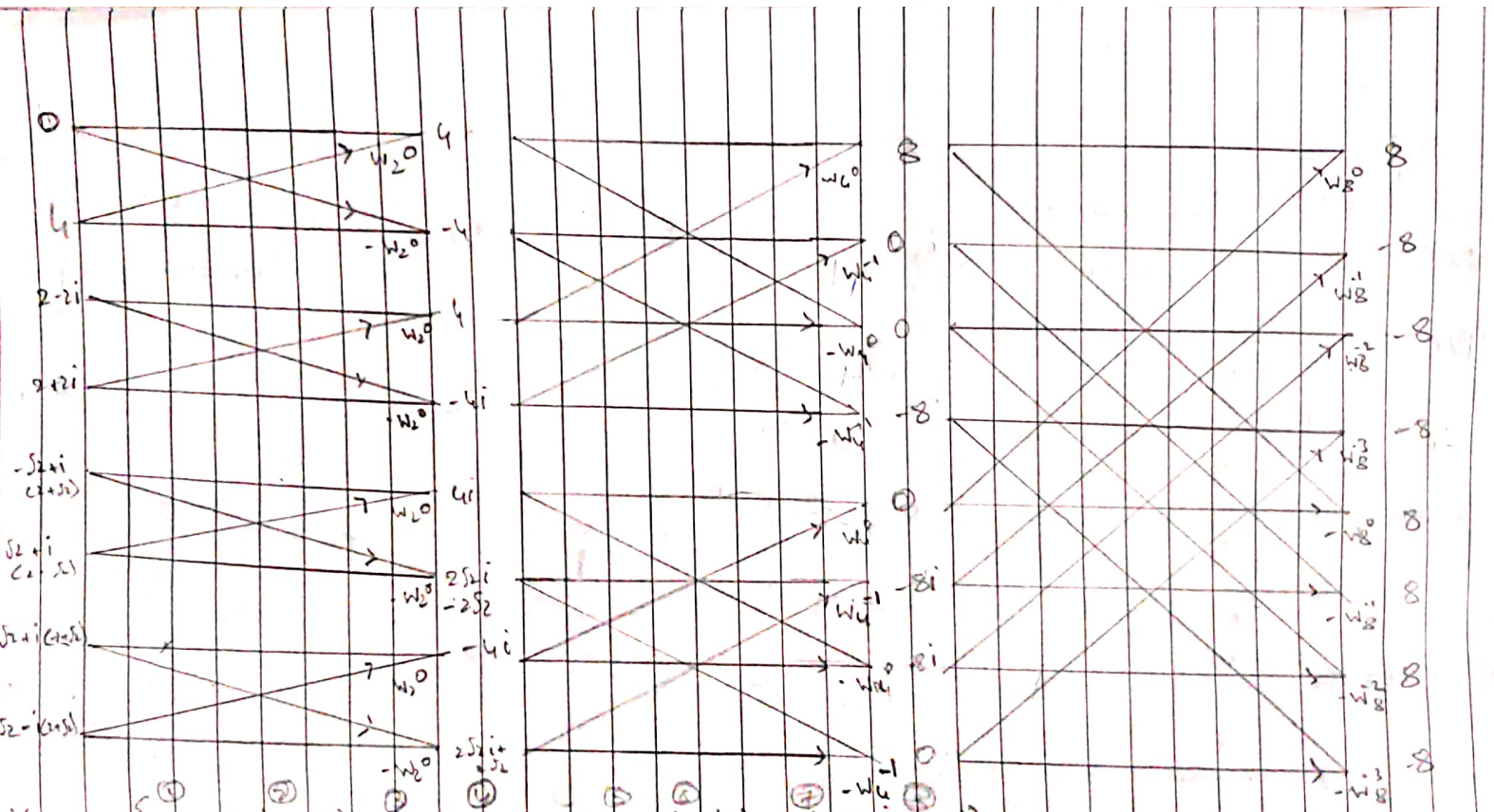


$\{1,2,3,4,4,3,2,1\}$

$\{1,3,4,2\}$ $\{2,4,3,1\}$ $\Rightarrow \{1,4\}$ $\{3,2\}$ $\{2,3\}$ $\{4,1\}$

FFT $= \{20, -3-2\sqrt{2}+i(-\sqrt{2}+1), 0, 2\sqrt{2}-3+i(1+\sqrt{2}), 0, 2\sqrt{2}-3-i(\sqrt{2}-1), 0, -3-2\sqrt{2}+i(1+\sqrt{2})\}$

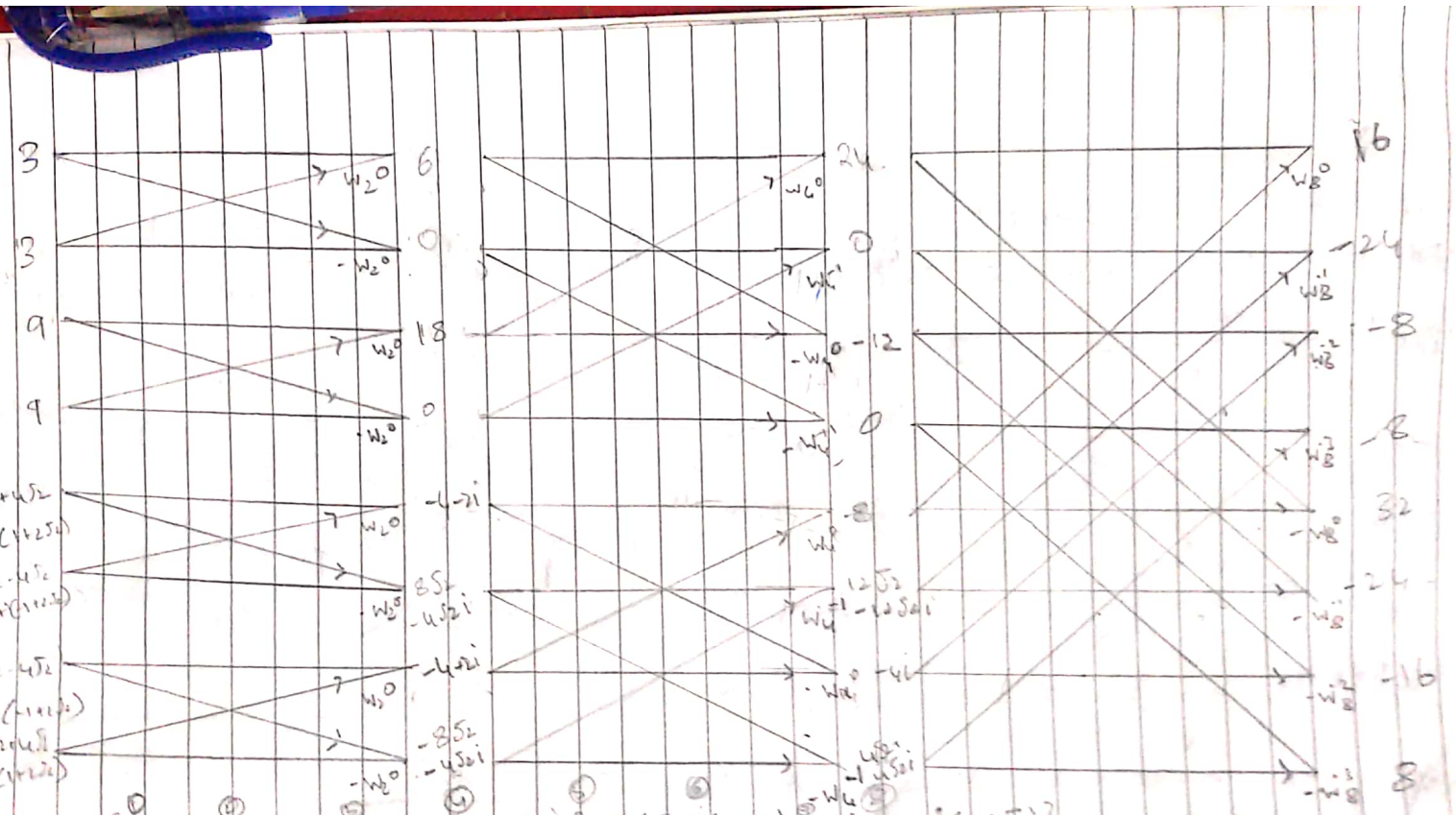
Scanned with CamScanner



$X(k) = \{0, -\sqrt{2}+i(2+\sqrt{2}), 2-2i, \sqrt{2}+i(2+\sqrt{2}), 4, \sqrt{2}+i(2-\sqrt{2}), 2+2i, -\sqrt{2}+i(2+\sqrt{2})\}$

$x(k) = \{1, -1, -1, -1, 1, 1, 1, -1\}$

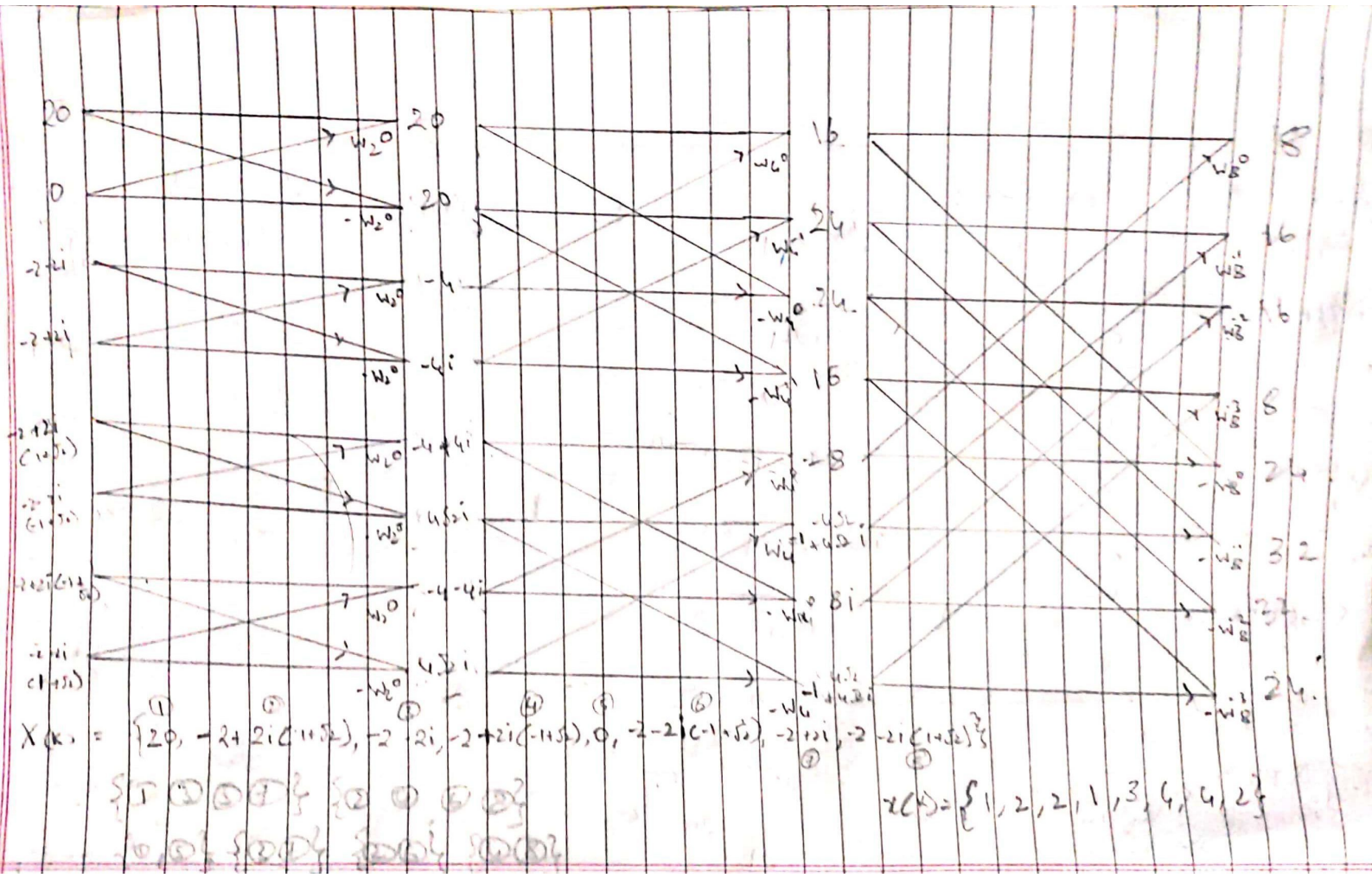
Scanned with CamScanner



$X(k) = \{3, -2+4\sqrt{2} - i(1+2\sqrt{2}), 9, -2-4\sqrt{2} - i(-1+2\sqrt{2}), 3, -2-4\sqrt{2} + i(-1+2\sqrt{2}), 9, -2+4\sqrt{2} + i(1+2\sqrt{2})\}$

$x(n) = \{2, -3, -1, -4, 4, 3, -2, 1\}$

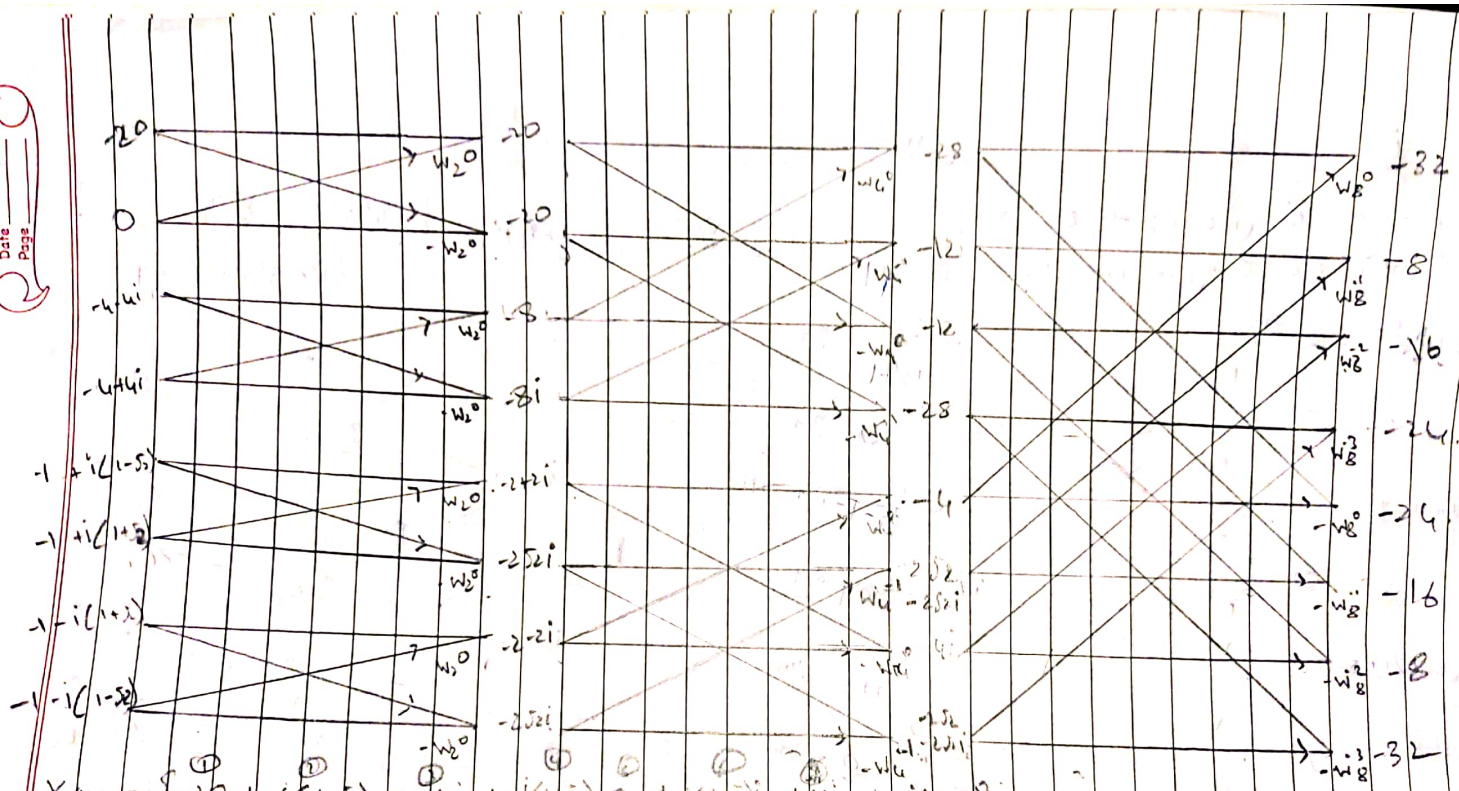
Scanned with CamScanner



$X(k) = \{20, -2+2i(1+\sqrt{2}), -2-2i, -2+2i(-1+\sqrt{2}), 0, -2-2i(-1+\sqrt{2}), -2+2i, -2-2i(1+\sqrt{2})\}$

$x(n) = \{1, 2, 2, 1, 3, 4, 4, 2\}$

Scanned with CamScanner



$X(k) = \{-20, -1+i(1-\sqrt{2}), -4-4i, -1-i(1+\sqrt{2}), 0, -1+i(1+\sqrt{2}), -4+4i, -1-i(1-\sqrt{2})\}$

$x(n) = \{-4, -2, -2, -3, -3, -2, -2, -4\}$